Projectile Motion Simulation

Situation:

1. One day after school, you are enjoying a soda in the backyard. When the can is empty, you decide to throw it in the trash can. What affects whether or not it gets in the can?

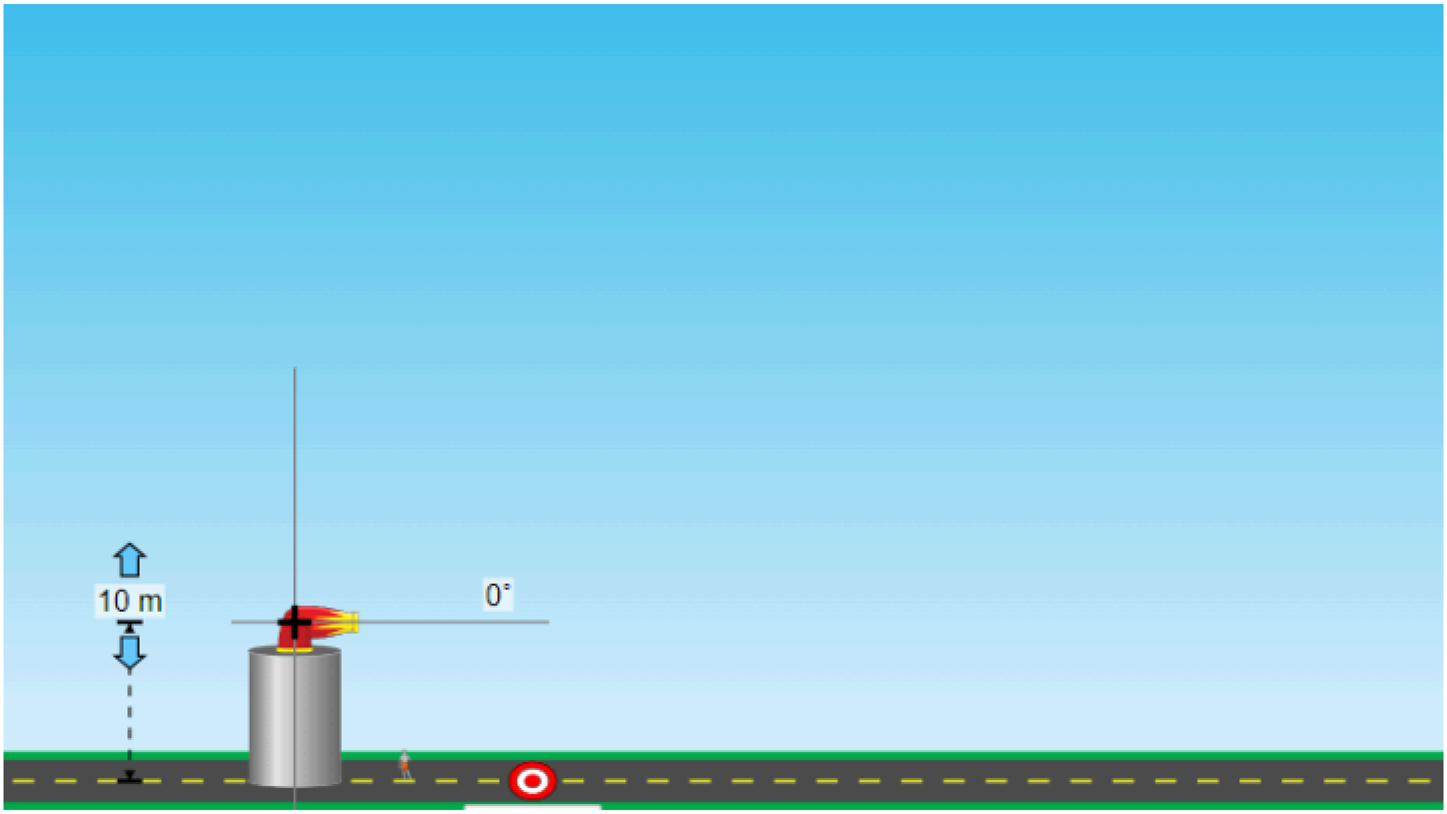
2. Play with the online simulation, *Projectile Motion*, to test your ideas about the things that affect the landing location of a projectile. Select the intro option on the 1st screen.

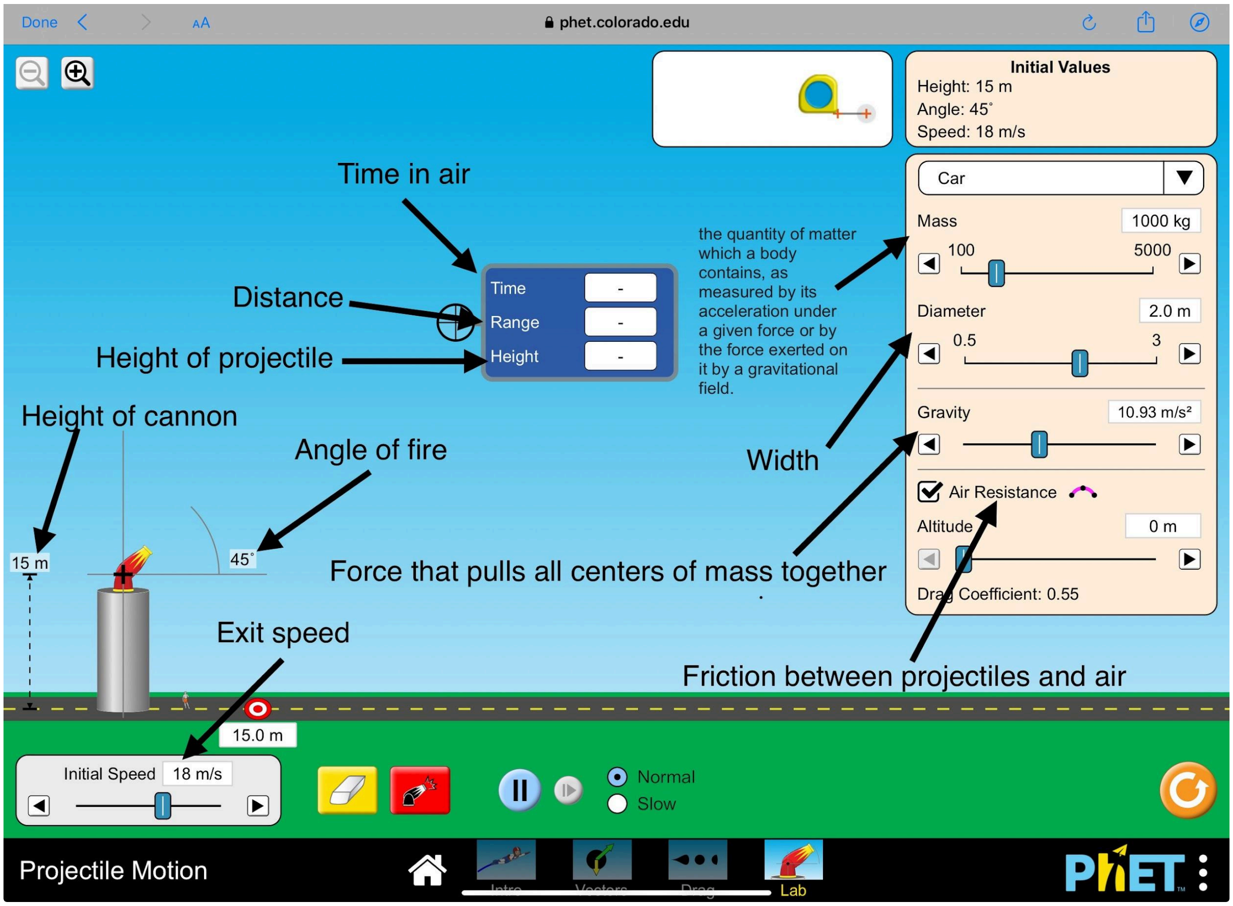
- Make a complete list of things that affect the landing site of a projectile and any discoveries you made using the simulation

- Next to each item above, briefly explain why you think the landing location changes (use physics terminology).

3. What is meant by the expression "flight path of a projectile"? Draw the flight path of your soda can and describe the shape. Use the simulation to investigate how the items you listed in #2 affect the shape of the flight path. Briefly summarize your discoveries, including explanations for the different flight paths for each of those items.

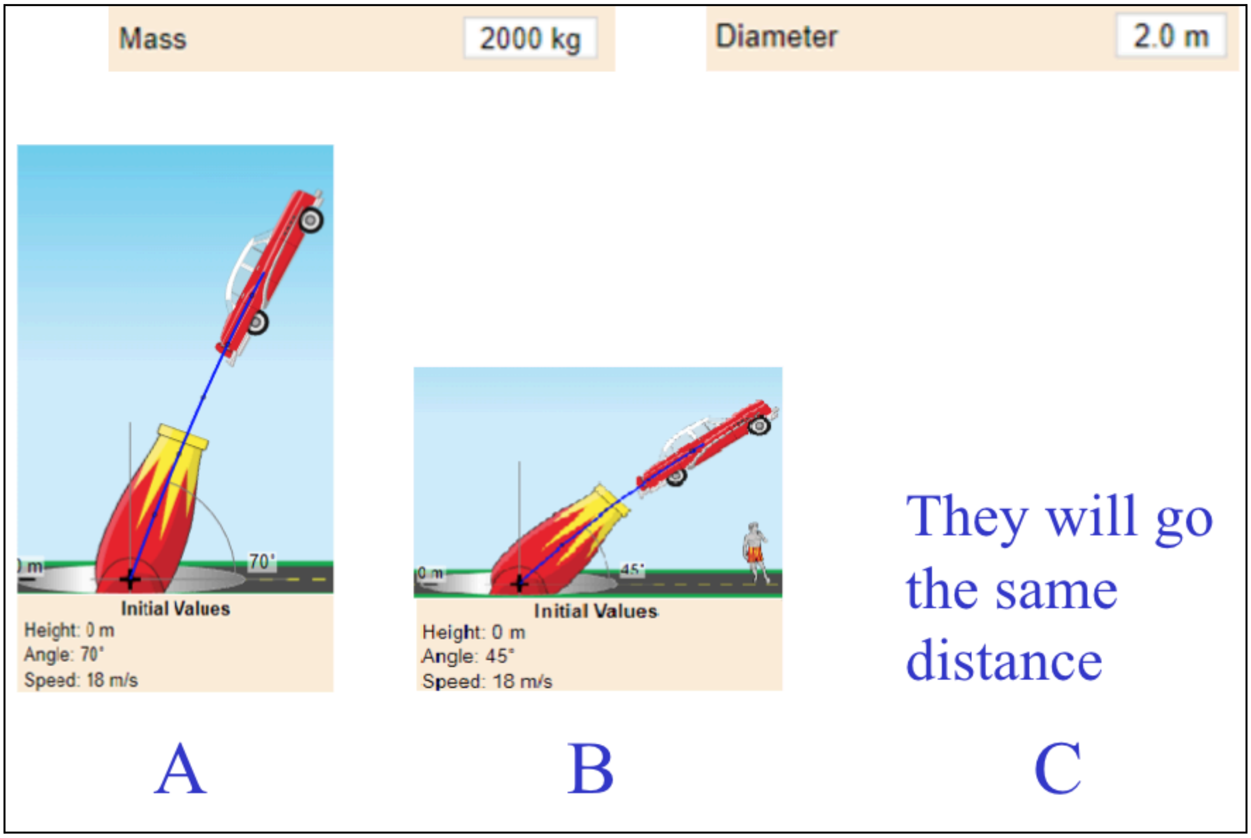
Flight path drawing:(double click on picture and add flight path)


10 m
0°



4. Describe why using the simulation is a good method for studying projectiles. Clearly identify error sources the simulation eliminates or minimizes that would be difficult to avoid in real-life. Also, run tests to determine how well the simulation represents projectile motion and identify limitations.

Test your understanding and self-check: For each question that follows, predict your answer first, then verify your answer by testing it. Support your answer with a possible explanation for the outcome you saw in the simulation.



1. Which launched car will (a) go **farther**, (b) be **in the air longer**, (c) **go higher**? Assume that both cars have the same gravity and there is no air resistance.

   a) Prediction:___ Correct Answer:______

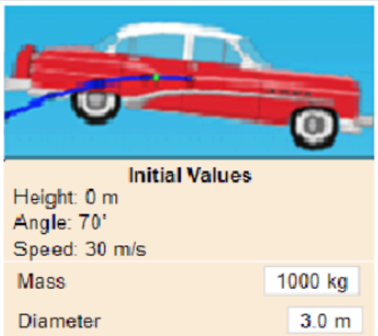
   Rationale:

   b) Prediction:___ Correct Answer:______

   Rationale:

c) Prediction:___ Correct Answer:______

Rationale:



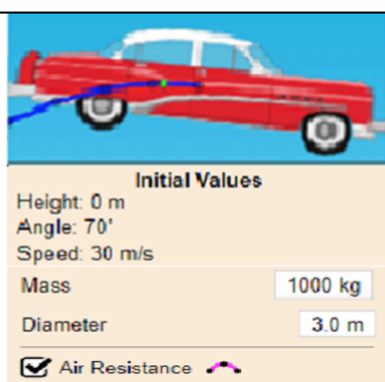
They will go same distance

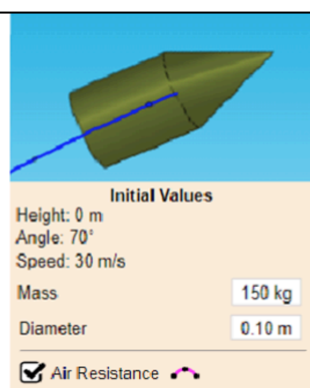
A B C

2. In the scenario shown above, which car will go **farther** if both have the same initial conditions, gravity and **without air resistance**?

Prediction:___ Correct Answer:______

Rationale:





They will go same distance

A B C

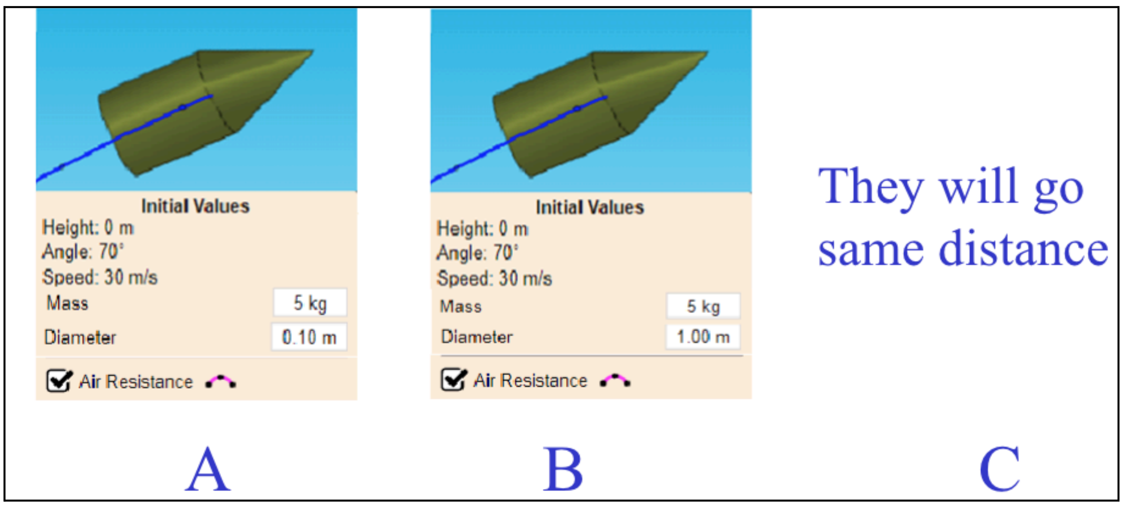
3. In the scenario shown above, which car will go (a) **farther**, (b) **higher** if both have the same initial conditions, gravity and **<u>with</u> air resistance**?

    a) Prediction:___   Correct Answer:______

    Rationale:

    b) Prediction:___   Correct Answer:______

    Rationale:



4. In the scenario shown above, which car will **go farther** if they both have the same initial conditions and air resistance with **different diameters**?

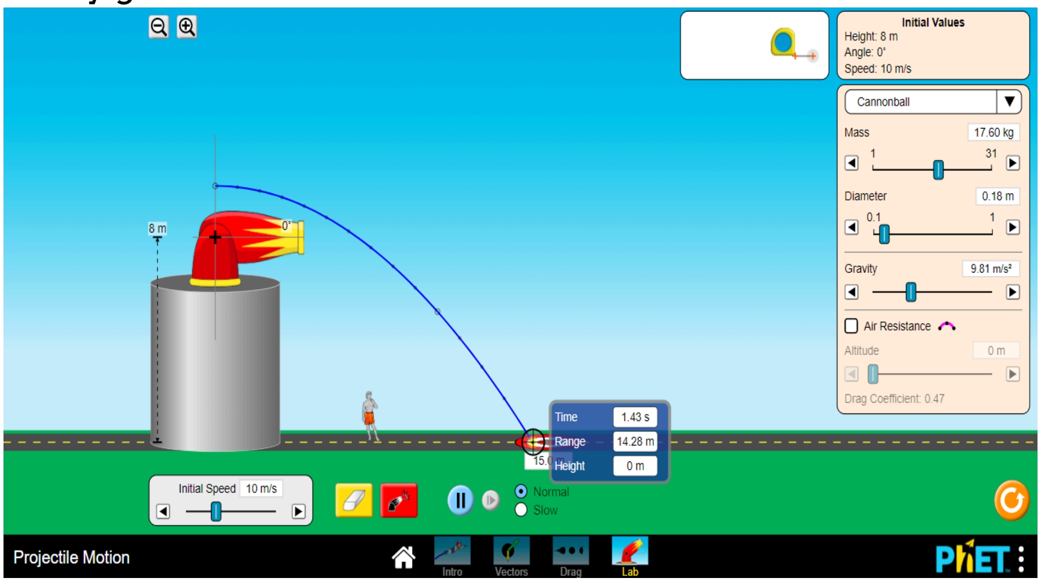
    Prediction:___   Correct Answer:______

    Rationale:

Horizontal Projectile Motion

Experiment Procedures:

1. Set the height of the launcher at 10 m and the angle $\theta = 0^{\circ}$
2. Set the velocity $(v_1 = 5 m/s)$, $(v_2 = 15 m/s)$, $(v_3 = 25 m/s)$ and find the time for each velocity given.



3. Record your data in the table below

Velocity (m/s)	Time (s)
5 m/s	
15 m/s	
25 m/s	

Questions

1) What do you conclude from the table above?

2) **If air resistance was present, does it affect the range of the projectile? If yes, explain your result.**

3) **Does increasing the diameter affect the time and range during the presence of air resistance? Give examples.**

Conclusion of this experiment:(what did you figure out?)